



β -Lactam Resistance Gene NDM-1 in the Aquatic Environment: A Review

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Abstract

New Delhi Metallo- β -lactamase-1 (NDM-1) offers carbapenem antibiotics resistance that creates an evolving challenge in treating bacterial infections. NDM-1-bearing strains were observed in surface waters around New Delhi in 2010 and after then identified globally. The usage of antibiotics may hasten the growth of the NDM-1-producing bacteria, which pose severe hazards to human and animal health. The emergence of the NDM-1 in the aquatic environment is turning out to be a growing concern worldwide. NDM-1 gene conferring resistance to a widespread class of antibiotics has been observed in bacteria disseminated in animal production wastewaters, hospital sewage, domestic sewage, industrial effluents, wastewater treatment plants, drinking water, surface water, and even in groundwater. This review recapitulates the currently published research studies on the prevalence and geographical distribution of the NDM-1 gene in the aquatic environment, its habitats, and healthcare risk associated with NDM-1-producing bacteria, in addition to molecular techniques employed to reveal the occurrence of the NDM-1 in the aquatic environment, including conventional polymerase chain reaction, real-time qPCR, DNA hybridization, and microarray-based methods.

Introduction

NDM-1 (New Delhi Metallo- β -lactamase-1), a recently evolving carbapenem, hydrolyzes a broad range of β -lactams antibiotics, including carbapenems [1]. The accelerated distribution of NDM-1-bearing strains poses a significant hazard to human health worldwide [2]. In recent times, resistant bacteria bearing NDM-1 genes were observed in surface water samples in New Delhi, India [3]. Intrinsically, exposure of water-borne NDM-1 was mediated via polluted urban water. The study on surface water or potable water contamination with bacteria harboring NDM-1 has also been described [3]. Since these primary findings, NDM-1 genes have now been observed in different places worldwide [4]. Although, significant pathways for extensive dissemination of the NDM-1 are not apparent, while international voyage and travel have been concerned [5]. Initially, most described positive isolates of NDM-1 were discovered in India, Pakistan, and Bangladesh [6]. Nevertheless, several clinical

infection cases induced by the NDM-1-carrying bacteria have been delineated in several other nations. It includes the USA [7], Canada [8], New Zealand [9], Germany [10], Switzerland [11], France [12], Australia [13], China [14], Japan [15], and Egypt [16], which foretell the global propagation of the NDM-1 genes. Ominously, NDM-1 genes have been observed in several compartments of the environment, comprising hospital sewage [17], tap water [3, 18], seepages, and municipal wastewaters [19]. It indicates the hazard that NDM-1-producing bacteria could broadly distribute the NDM-1 gene from medicinal lineage. The environmental existence of NDM-1 genes has severe consequences for public health. The existence of the NDM-1 in water and sediment samples causes a severe threat to citizen's dependent on public water supply and sanitation amenities [3]. The previous study affirmed that NDM-1 genes emerged in and persisted in the disinfecting operation of wastewater treatment plants (WWTPs) [19]. The alleviation of the NDM-1 gene proliferates the prospect of NDM-1 horizontal transfer. The earlier study demonstrated the transmission of the NDM-1 gene to the river's aboriginal bacteria [19]. It enhances the level of the distribution of the NDM-1 genes into the environmental setting. Additionally, WWTPs expedite the transmission of the NDM-1 genes from clinical pathogens to environmental microbes, as receiving water is

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used by human for various activities [20, 21]. This review paper outlines the state-of-the-art knowledge existing in the published research on the existence and geographical distribution of the NDM-1 gene in the water environment, its habitats, healthcare risk associated with NDM-1-producing bacteria, and molecular methods to detect the environmental occurrence of the NDM-1.

Geographical Distribution of the NDM-1 in the Water Environment

NDM-1 was originated in the year 2008 in a Swedish citizen hospitalized in New Delhi [22]. It has been anticipated that India and its subcontinent are the crucial pool of NDM-1-carrying bacteria depending on the number of fatalities caused by NDM-1-bearing bacteria in several worlds [23]. The geographical distribution of the NDM-1 gene has been designated in the published literature and occurrence on the gene worldwide (Fig. 1).

Asian continent constitutes the primary reservoir with around 55.10% abundance of NDM-1 gene disseminated typically in India and China, with low prevalence in Pakistan, Bangladesh, Myanmar, Vietnam, the Philippines, and Saudi Arabia. Table 1 summarizes the information on the prevalence of the NDM-1 gene across the globe. Europe exhibits around 18.36% abundance of the NDM-1 gene distributed in the United Kingdom, Serbia, Switzerland, Ireland, Sweden, Spain, Belgium, and the Czech Republic. North and South American continent combinedly display around 8.16% occurrence of the total NDM-1 gene as reported worldwide, principally spread in the USA and

Canada. Africa accounts for 14.28% reservoir of the total NDM-1 gene dispersed worldwide. Algeria and Nigeria displayed the leading distribution, whereas South Africa and the Democratic Republic of the Congo demonstrated a low prevalence of the African subcontinent's NDM-1 gene. The Arctic region showed 4.08% of the total pool of the NDM-1 gene.

Habitats of NDM-1 in the Water Environment

Effluents from domestic sewage, industrial waste, urban/agricultural runoff, and hospital in emerging nations signify a substantial source of emerging pollutants, for example, antibiotic-resistant bacteria (ARB) and antibiotic-resistant genes (ARGs), and metals for the receiving water vicinities. The industrial effluents and domestic sewage are released to the sewer, lakes, rivers, and seas, lacking a proper degree of treatment that may have accrued in the sediments [71, 72]. Lakes and rivers are contemplated as presumed pools of emergent contaminants (ARGs, metals, drugs). They receive wastewaters comprising several pollutants from diverse sources [73]. Reports inspecting rivers as a prospective ARG's reservoir are mostly restricted in tropical zones comprising South India [74]. However, some reports have described clinically significant ARB and ARGs in the environment from the Northern part of India [3, 24]. Consequently, the widespread usage of human and veterinary antibiotics in a clinical setting, hospital sewage, and farm animal manure is treated as the primary source of environmental ARGs. It can flow into the aquatic environment by directly releasing domestic sewage or into sewage treatment



Fig. 1 Distribution of the NDM-1 gene in geographically isolated water environments

Table 1 NDM-1 gene in the aquatic environment

Continent/country	Environmental source	Reference
Asia/New Delhi, India	Seepage and tap water samples	[3]
Asia/India	Water and sediment samples from Yamuna River in Delhi and the Ganga River near Rishikesh-Haridwar	[24]
Asia/Chennai, India	Chlorinated and purified drinking water samples before public distribution and water samples from the lake	[25]
Asia/Tiruchirappalli, India	Water and sediment samples from the Cauvery River Basin	[26]
Asia/Aligarh, India	Hospital sewage samples	[27]
Asia/Pune, India	Sediment samples from Mutha River	[28]
Asia/New Delhi, India	Water and sludge samples from different unit operations in Sewage Treatment Plant (STP)	[29]
Asia/New Delhi, India	Drain sites, hospital waste outfalls, STP effluent, and water samples from Yamuna River	[30]
Asia/Kerala, India	Soil and water samples from water bodies adjacent to hospitals	[31]
Asia/Beijing, China	Water samples from the river, treated drinking water samples, raw water samples from the pools, and sewage from hospitals and community	[32]
Asia/China	Water and sludge samples from the outlet of each unit of municipal WWTPs	[33]
Asia/Hangzhou, China	Soil samples surrounding animal farms in Zhejiang province	[34]
Asia/Nanjing, China	Source water, the effluent of sedimentation tank, filtered water, chlorine-disinfected water, and tap water samples	[35]
Asia/China	WWTP effluent and water samples from the receiving river	[36]
Asia/China	Surface water and sediment samples from Tai Lake	[37]
Asia/Chengdu, China	Hospital sewage samples from the influx mainstream of the WWTPs at West China Hospital	[38]
Asia/Weifang, China	Drinking water samples from backyard animal farms	[39]
Asia/China	Samples from source water and water distribution system	[40]
Asia/China	Drinking water supply systems	[41]
Asia/China	Hospital sewage	[42]
Asia/China	Water samples from natural water bodies: Poyang Lake and Haihe River	[43]
Asia/Central Hanoi, Vietnam	Seepage water samples from a water pool, lakes, and rivers	[44]
Asia/Pakistan	Water samples from Islamabad, Rawalpindi, and Khyber Pakhtunkhwa	[45]
Asia/Jeddah, Saudi Arabia	Untreated wastewater sample from a WWTP	[46]
Asia/Dhaka, Bangladesh	Tap water samples and wastewater samples from an open drainage system	[47]
Asia/Yangon, Myanmar	Sewage and drinking water samples	[48]
Asia/Philippines	Hospital sewage and river water samples	[49]
Europe/Cardiff, The United Kingdom	Sewage effluent samples from the WWTP	[3]
Europe/Switzerland	Water samples from lakes and rivers	[50]
Europe/Belgrade, Serbia	Water samples from lakes, rivers, and springheads	[51]
Europe/Ireland	Recreational water and sewage samples	[52]
Europe/Spain	Water samples from the Ter River	[53]
Europe/Spain	Samples from hospital wastewater, raw sewage, and STP effluent. Sediment and bio-film samples from Montsant River, Tarragona, and Matarranya River, Teruel	[54]
Europe/Örebro, Sweden	Water samples from Svartån River and Hjälmaren Lake	[55]
Europe/Brussels, Belgium	Water samples from the hospital, raw, and treated wastewater samples from WWTPs	[56]
Europe/The Czech Republic	Municipal WWTPs	[57]
North America/Manitoba, Canada	Drinking water and source samples from the community in the Island Lake Region	[58]
North America/Washington, The USA	Water samples from municipal WWTPs, irrigation canals, and rivers	[59]
North America/New Jersey, The USA	Sewer sediments and post-screen wastewater influent samples	[60]
South America/Brazil	Coastal water	[61]
Africa/Algeria	Hospital sewage	[62]
Africa/Béjaïa, Algeria	Soil and water samples	[63]
Africa/Tunisia	Wastewater samples from hospitals	[64]

Table 1 (continued)

Continent/country	Environmental source	Reference
Africa/Lagos, Nigeria	Water samples from recreational beaches	[65]
Africa/Nigeria	Water and soil samples	[66]
Africa/Eastern Cape Province, South Africa	Surface water, STP effluents, hospital effluents, irrigation water, and soil samples	[67]
Africa/Kinshasa, Congo	Wastewater and sediment samples from the river	[68]
Arctic region	Arctic/sub-Arctic(polar) marine sediment samples	[69]
Arctic region/Kongsfjorden	Soil samples	[70]

plants (STPs) through sewage systems and then discharge of effluents and sludge into the environment [75]. ARB and ARGs might potentially be transmitted into soils by altering farmland with animal dung and treated bio sludge from STPs. Therefore, they can percolate to groundwater or be transported by surface water and runoff [76]. Surface water and groundwater are typically exploited as potable water sources; therefore, ARBs and ARGs can enter through water treatment systems and infiltrate into water supply networks [77].

Hospital Sewage

The commonly used antibiotics for human, animal, and aquaculture purposes may employ selection pressure on bacteria in clinical settings [78], livestock farming [79], and aquaculture [80]. Therefore, they are believed to be primary sources accountable for the distribution of ARGs into the environment. Several studies described the presence of NDM-1 gene from hospital sewage outlets [26, 27], wastewater effluent of 12 hospitals [30], water bodies adjoining hospitals situated in Kollam, Ernakulum, and Kannur districts of Kerala [31], sewage from four comprehensive hospitals from Beijing City [32], hospital sewage collected from the inlet of the WWTP at West China Hospital [38], hospital sewage [42, 62], hospital wastewater effluent [54], water samples from the hospital [56], wastewater samples from seven selected hospitals [64], hospital effluents [67], and sewage sample from seven hospitals [49] as reported in Table 1.

Wastewater from Aquaculture and Animal Farms

A few studies also reported the abundance of NDM-1 from five aquaculture farms from Kerala [31], soil samples surrounding livestock farms in Zhejiang province [34] and water samples from backyard animal farms [39]. These wastewaters containing the NDM-1 gene are rapidly released to the environment that ultimately is carried to the adjacent lakes, rivers, streams, or other water bodies or percolate underneath via soil through precipitation.

Sewage Treatment Plant (STP) and Wastewater Treatment Plant (WWTP) Effluent

STP plays an essential function in the exchange and propagation of ARGs in the environment. They receive the ARB with the influx of sewage originated from the domestic household, hospitals, industry, and agriculture [81]. Sludge applications on agricultural lands and STP effluent are identified as vital sources of ARB and ARGs to the surface water, soil, and then into groundwater [76]. Numerous studies showed that bacteria carrying ARGs could be discharged into surface water from STPs [75, 82, 83]. Several previous studies [29, 30, 54, 67] revealed that STPs act as vital reservoirs for the NDM-1 gene (Table 1). Thus, STPs are an essential interface between diverse water bodies, including domestic wastewater, surface water, groundwater, and hospital wastewater. Hence, they may expedite the exchange and distribution of genes between these compartments of the environment [84].

WWTPs are considered a critical habitat for the acquisition and dissemination of ARB into the water environment. There are three significant causes set forth to withstand this notion based on several studies: (1) the persistent discharge of antibiotics, ARB, and ARGs accumulated in the hospital and municipal sewerage systems; (2) the favorable environmental situations for transmission of ARGs among numerous bacterial genera in WWTPs, and (3) elevated concentrations of various ARGs offering resistance against commonly used drugs and antibiotics from WWTP final effluents [56]. For quite a few years, most emerging nations use traditional WWTPs, and adjoining water bodies act as accepting reservoirs from manufacturing industries and hospitals directly [85]. Several prior studies [3, 36, 46, 56, 59] have shown that WWTP serves as important reservoirs for the NDM-1 gene (Table 1). In addition, several hospitals discharge effluents directly into water bodies without a proper degree of treatment, enhancing the probability of identifying ARGs. This unsanitary process has led to the prevalence and distribution of ARB and ARGs and a severe human health risk global scale. A few studies have reported hospital wastewater as a significant source of ARGs, although conventional WWTPs cannot remove ARB.

Water Bodies

Natural water bodies comprise lakes, canals, rivers, seas, and oceans. Water is utilized to irrigate plants consumed by humans and animals, contaminating crops that could also direct human/animal accumulation with ARB [86]. Recently, the NDM-1 genes were broadly spread in several different species of bacteria in water sources. The high predominance of NDM-1 gene-carrying bacteria in India's municipal drinking water supply is a matter of concern [3]. This particular gene is of significant concern to public health as it offers resistance to all β -lactams groups of antibiotics [3]. ARGs have been observed in the microbial community's isolates in the water bodies, including lakes, streams, and rivers that were unpolluted or marginally contaminated [87–89]. Numerous studies (Table 1) reported the existence of the NDM-1 gene in natural water bodies [24–26, 30, 32, 40, 43, 44, 49–52, 54, 55, 65, 68] and have shown ARGs in aquatic sediments, mainly those sediments exposed to human activities [90, 91]. In addition, various prior studies (Table 1) demonstrated the NDM-1 presence in sediment samples [24, 26, 28, 37].

Drinking Water

Several ARG obtained from the drinking water supply network has been depicted in recent times [92, 93]. Numerous studies [25, 32, 35, 39, 40, 58] showed the occurrence of the NDM-1 gene in drinking and tap water samples (Table 1).

Significant Healthcare Risk Associated with NDM-1-Carrying Bacteria

NDM-1-bearing strains are specifically dangerous because (a) the transferability and capability of most plasmids found in these bacteria lead to a widespread horizontal transfer among the bacterial community; (b) the absence of a usual standardized phenotypic test for NDM-1 producers; (c) high occurrence of unrecognized asymptomatic bacterial strains apparently; (d) lack of effective antibiotics accessible for treating illness caused by NDM-1-carrying bacteria [94]. Infections induced by NDM-1-producing

bacteria comprise diarrhea, peritonitis, septicemia, and infections of soft tissue, lungs, and urinary tract. They infect the host by affecting the blood, lungs, wounds, and urinary tract [16]. The means of transfer of NDM-1-producing bacteria could either be via body fluids or through cross-contamination during cooking and may appear in the hospital setting or the community [95].

Molecular Methods Used for the Screening and Characterization of the NDM-1

There is a need to develop and employ a sophisticated molecular technique to analyze the existence, fate, and transmission of the NDM-1 in the environment by considering that the NDM-1 is widespread in water environments cited above. Until now, the techniques applied for screening and characterization of NDM-1 have included, but have not been restricted to, conventional polymerase chain reaction (PCR), DNA sequencing, quantitative PCR (qPCR), and DNA hybridization methods comprising microarray [85]. Conventional PCR analysis has commonly been applied for the detection of environmental ARG. The qPCR is frequently applied for the quantification of target DNA. It is based on the principle that the number of products obtained during each amplification cycle of the qPCR is directly proportional to the template before the start of the PCR analysis [96]. SYBR Green-based method is the most commonly used for quantifying ARGs among the various fluorescent dye developed for qPCR analysis. DNA microarray is a genome-enabled technology that offers rapid, high-throughput, and higher precision compared to other molecular techniques to detect ARGs. DNA microarray can offer clinically significant knowledge on the bacterial isolates in depth by identifying several ARGs' existence or non-existence concurrently in a specific assay [97]. However, the use of microarray techniques in most environmental samples is quite limited due to the samples' pretreatment and complexity. Several previous studies were reported using these molecular techniques to detect the NDM-1 in the water environment (Table 2).

Table 2 The molecular technique for the screening and characterization of the NDM-1

Molecular techniques	Reference
Conventional PCR	[3, 25, 27, 28, 31–34, 38, 43–45, 47, 49–51, 59, 63, 66, 67]
qPCR	[24, 26, 29, 30, 36, 37, 40, 43, 46, 52–54, 56, 58, 60, 62, 64, 65, 68–70]
Metagenomics	[35, 40, 60]
Whole-genome sequencing and oligonucleotide microarray	[38–40, 42, 46, 48, 49]

Drivers of the Emergence of the NDM-1 Worldwide

Several factors that influence the occurrence of NDM-1 include the usage of antibiotics, deficiency of clean water and sanitation facilities, travel, and trade. Worldwide use of antibiotics has increased in recent times; however, precise data are not accessible. The study conducted by [98] revealed the increased antibiotic use in India as India manufactures at least one-third of the total oral and hypodermal antibiotics produced in the world. Total antibiotics consumption increased twofold between 1999 and 2008 in a tertiary treatment hospital in Delhi, India [99]. In several parts of the globe, antibiotics can be bought without a valid prescription [98]. They are also consumed broadly for some commercial crops, in food animals, and aquaculture. The presence of antibiotic residues in domestic sewage, soil, industrial effluents, and surface water, including lakes and rivers, exerts intense selective antibiotic pressures on the global microbiome over the past 6–7 decades, unparalleled in human history [100]. Regions with insufficient sanitary facilities are especially susceptible to the distribution of ARGs in fecal flora, leading to water contamination and food contamination. The worldwide movement of bacteria and resistance genes is also facilitated by travel and tourism. The earlier study reported the global dissemination of a widespread plasmid that offered resistance to aminoglycosides [101]. Travel can drive to bacterial flora changes and ARGs acquisition, even in the non-existence of infection. Thus, tourists perform a significant role in transmitting NDM-1-producing bacteria to several nations.

The Future Perspective

NDM-1 gene-producing bacteria are now distributed broadly across the globe (Fig. 1). Their ability to transmit this plasmid-mediated resistance gene to numerous bacteria species and its association with several other ARGs will cause enormous clinical treatment difficulties. However, studies are ongoing to evaluate different combinations of drugs that may effectively treat bacterial infection. In the near future, the emphasis must be on supplying clean drinking water and elementary sanitation amenities in the regions where the environment is previously polluted or at a greater risk of contamination with NDM-1-carrying bacteria. It should also focus on detecting those who transmit these bacteria, restricting antibiotics' abuse, and inhibiting distribution through infection control. The strategies have to be formulated to diminish

the likelihood of NDM-1 gene entry and their propagation in the environment. The most effective method is supposed to be the judicious usage of antibiotics in the clinical setting and agriculture. Novel and efficient wastewater treatment methods should be designed to enhance the removal efficiency of the NDM-1 gene in STPs. Furthermore, the viability of agricultural use of sludge or reclaimed wastewater in irrigation has to be conferred carefully as the potential release of the NDM-1 gene to groundwater and soil. Studies on transmission and degradation pathways of the NDM-1 gene and their health risk assessment should be performed to control the distribution of the environmental NDM-1. Unfortunately, although the instantaneous distribution of NDM-1 genes is going on, information is still deficient for epidemiological analysis. It is significant to ascertain situations for estimating the propagation of NDM-1 from the environment to clinically significant pathogens.

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Declarations

Conflict of interest The authors declare that they have no known competing financial or personal interests.

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